Dominance and common deformations of short local rings

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September 9, 2026

Abstract

Let $(R, \mathfrak{m}, k)$ be a commutative Artinian local ring with $\mathfrak{m}^3 = 0$. We construct a one-dimensional complete Cohen–Macaulay local ring whose quotients by two regular elements outside the square of its maximal ideal are R and $\operatorname{gr}_{\mathfrak{m}} R$. Consequently, dominance and uniform dominance are each equivalent for R and its associated graded ring. Their dominant indices differ by at most the bounds $\operatorname{dx}(R) \leq 4 \operatorname{dx}(\operatorname{gr}_{\mathfrak{m}} R) + 3$ and the reverse inequality when finite. Combining this construction with Kimura's equicharacteristic dominance theorem and known deformation and flat-descent results, we prove that every non-complete-intersection Gorenstein local ring satisfying either $\mathfrak{m}^3 = 0$ or $e(R) \leq \operatorname{codim} R + 2$ is dominant. This answers both clauses of Kobayashi–Takahashi Question 7.1 without restrictions on characteristic or residue-field cardinality.

1 Introduction

All rings in this paper are commutative, Noetherian, and have identity. A local ring $(R, \mathfrak{m}, k)$ is *dominant* if

$$k \in \operatorname{thick}_{\mathsf{D}_{\mathrm{sg}}(R)}(X) \qquad \text{for every nonzero } X \in \mathsf{D}_{\mathrm{sg}}(R).$$

Here $\mathsf{D}_{\mathrm{sg}}(R)$ is the quotient of the bounded derived category of finitely generated modules by the perfect complexes. A thick subcategory is closed under shifts, cones, and direct summands. Dominance was introduced by Takahashi in connection with subcategory classification [6]. For an Artinian ring, k generates $\mathsf{D}_{\mathrm{sg}}(R)$, so dominance says that this category has no nonzero proper thick subcategory.

Short local rings, here meaning those with $\mathfrak{m}^3 = 0$, have long provided examples and tests for homological conjectures; see Avramov–Iyengar–Şega [1]. In the Gorenstein case, Kobayashi and Takahashi ask whether every non-complete-intersection ring with $\mathfrak{m}^3 = 0$, or more generally with $e(R) \leq \operatorname{codim} R + 2$, is dominant [3, Question 7.1]. The multiplicity is Hilbert–Samuel multiplicity, and $\operatorname{codim} R = \operatorname{edim} R - \dim R$.

Kimura recently proved the short case when R contains a field, and the higher-dimensional case in equal characteristic with infinite residue field [2, Theorem 7.3]. His theorem supplies the essential representation-theoretic input here. In particular, the first non-complete-intersection Gorenstein dominant examples already belong to that work. Our contribution is a common deformation that transfers dominance from the associated graded ring to an arbitrary short local ring, including rings without a coefficient field. The construction does not require the Gorenstein hypothesis.

Theorem 1.1. *Let $(R, \mathfrak{m}, k)$ be an Artinian local ring with $\mathfrak{m}^3 = 0$, and put $G = \operatorname{gr}_{\mathfrak{m}} R$. There exist a one-dimensional complete Cohen–Macaulay local ring $(T, \mathfrak{n}, k)$ and T-regular elements $u, v \in \mathfrak{n} \setminus \mathfrak{n}^2$ such that*

$$T/(u) \cong G, \qquad T/(v) \cong R. \tag{1}$$

If R has residue characteristic $p > 0$ and $p \neq 0$ in R, one can take T finite free of rank $\ell_R(R)$ over a complete Cohen discrete valuation ring V with uniformizer p, with $u = p$ and $v = p - a$ for an element a satisfying $a^3 = 0$.

The elements u and v need not be distinct. The condition that both lie outside $\mathfrak{n}^2$ is essential for the categorical application: dominance need not descend through a regular element in $\mathfrak{n}^2$; see [6, Corollary 6.12].

Theorem 1.2. *With R and G as above, R is dominant if and only if G is dominant. Also, R is uniformly dominant if and only if G is uniformly dominant. If their dominant indices are finite, then*

$$\mathrm{dx}(R) \leq 4\,\mathrm{dx}(G) + 3, \qquad \mathrm{dx}(G) \leq 4\,\mathrm{dx}(R) + 3. \tag{2}$$

Theorem 1.3. *Let $(R, \mathfrak{m}, k)$ be a Gorenstein local ring that is not a complete intersection. If either $\mathfrak{m}^3 = 0$ or $e(R) \leq \operatorname{codim} R + 2$, then R is dominant.*

Thus Theorem 1.3 answers Question 7.1 of [3]. The removal of the characteristic restriction uses Theorem 1.1; the removal of the finite-residue-field obstruction in positive dimension uses Liu's flat descent [4]. We make no assertion that the rings in Theorem 1.3 are uniformly dominant. Theorem 1.2 transfers that stronger question to their graded Artinian counterparts without answering it.

2 The common deformation

Write

$$e = \dim_k(\mathfrak{m}/\mathfrak{m}^2), \qquad f = \dim_k \mathfrak{m}^2, \qquad N = 1 + e + f = \ell_R(R).$$

Since $\mathfrak{m}^3 = 0$, multiplication gives a well-defined symmetric bilinear map

$$b : (\mathfrak{m}/\mathfrak{m}^2) \times (\mathfrak{m}/\mathfrak{m}^2) \longrightarrow \mathfrak{m}^2. \tag{3}$$

Its values span $\mathfrak{m}^2$. The graded ring G is precisely the vector space $k \oplus k^e \oplus k^f$ with multiplication given by b in positive degree and all products of total degree at least three zero.

Lemma 2.1. *If R contains a field, then $R \cong G$ as local rings.*

Proof. The Artinian ring R is complete, so the equicharacteristic Cohen structure theorem gives a coefficient field identified with k; see [5, Tag 032A]. Choose a k-linear complement E of $\mathfrak{m}^2$ in $\mathfrak{m}$. The decomposition $R = k \oplus E \oplus \mathfrak{m}^2$ has exactly the multiplication described in (3), because $E\mathfrak{m}^2 = (\mathfrak{m}^2)^2 = 0$. Identifying E with $\mathfrak{m}/\mathfrak{m}^2$ gives the desired isomorphism. □

We now treat rings that do not contain a field. Their residue field has characteristic $p > 0$, with $p \neq 0$ in R. By the Cohen structure theorem there is a complete discrete valuation ring V of characteristic zero, with maximal ideal pV and residue field k, and a local homomorphism

$$V \longrightarrow R \tag{4}$$

inducing the chosen residue-field isomorphism [5, Tag 032A and its proof]. This statement holds for imperfect k as well. The map (4) is not asserted to be injective. All uses of scalars from V in R are through this map.

Choose elements $x_1, \ldots, x_e \in \mathfrak{m}$ whose classes are a basis of $\mathfrak{m}/\mathfrak{m}^2$, and a k-basis $z_1, \ldots, z_f$ of $\mathfrak{m}^2$. Define $b^t_{ij} \in k$ by

$$x_i x_j = \sum_{t=1}^{f} b^t_{ij} z_t.$$

This equality uses the canonical k-module structure on $\mathfrak{m}^2$, annihilated by $\mathfrak{m}$. Choose symmetric lifts $B^t_{ij} = B^t_{ji} \in V$ of these coefficients. Empty lists and empty sums are allowed when $f = 0$.

Define a free V-module

$$T = V \cdot 1 \;\oplus\; \bigoplus_{i=1}^{e} V X_i \;\oplus\; \bigoplus_{t=1}^{f} V Z_t \tag{5}$$

with a V-bilinear multiplication, unit 1, and rules

$$X_i X_j = \sum_{t=1}^{f} B_{ij}^{t} Z_t, \qquad X_i Z_t = 0, \qquad Z_s Z_t = 0. \tag{6}$$

These rules define an associative commutative algebra: on triples of nonunit basis elements, both associated products are zero; triples containing 1 satisfy associativity automatically. Put

$$J = \bigoplus_{i} V X_i \oplus \bigoplus_{t} V Z_t.$$

Then J is an ideal with $J^3 = 0$ and $T/J = V$. Therefore T is local, with maximal ideal $\mathfrak{n} = pT + J$, and has dimension one. It is Noetherian and complete: it is finite free over V, and the p-adic and $\mathfrak{n}$-adic topologies agree since $p^r T \subseteq \mathfrak{n}^r$ and $\mathfrak{n}^{r+2} \subseteq p^r T$. Multiplication by p is injective, so T is Cohen–Macaulay. Reducing (6) modulo p gives

$$T/pT \cong G. \tag{7}$$

The coefficient map and the assignments $X_i \mapsto x_i$, $Z_t \mapsto z_t$ define a ring homomorphism $\varphi : T \to R$. Indeed, the differences between the lifted coefficients and their residues annihilate $\mathfrak{m}^2$, and all products involving $\mathfrak{m}^2\mathfrak{m}$ vanish. It is surjective: first represent an element of R modulo $\mathfrak{m}$ by an element of V, then represent the remaining class modulo $\mathfrak{m}^2$ by a V-linear combination of the x_i, and finally represent the remainder by the z_t.

Adapt the bases to the nonzero element $p \in R$. If $p \notin \mathfrak{m}^2$, choose $x_1 = p$ and set $a = X_1$. If $p \in \mathfrak{m}^2$, choose $z_1 = p$ and set $a = Z_1$. In both cases $a^3 = 0$ and $\varphi(p - a) = 0$.

Lemma 2.2. *The element $p - a$ is T-regular, the map φ induces an isomorphism $T/(p - a) \cong R$, and both p and $p - a$ belong to $\mathfrak{n} \setminus \mathfrak{n}^2$.*

Proof. Let K be the fraction field of V. In $T \otimes_V K$, the element $p - a$ is invertible, with inverse

$$p^{-1}\bigl(1 + p^{-1}a + p^{-2}a^2\bigr).$$

Since T is V-free, it embeds in this algebra, proving that $p - a$ is regular.

In the ordered basis $(1, X_1, \dots, X_e, Z_1, \dots, Z_f)$, multiplication by a is strictly triangular. Thus multiplication by $p - a$ has determinant p^N. The length formula for the cokernel of a square matrix of nonzero determinant over a discrete valuation ring gives

$$\ell_V\bigl(T/(p - a)\bigr) = v_p(p^N) = N.$$

The R-composition factors of R are all k, which also has V-length one. Hence $\ell_V(R) = N$. The surjection $T/(p - a) \to R$ is an isomorphism by additivity of length.

Finally, the projection $\pi : T \to V$ with kernel J satisfies $\pi(\mathfrak{n}^2) = p^2 V$. Both p and $p - a$ project to $p \notin p^2 V$, so neither lies in $\mathfrak{n}^2$. □

Proof of Theorem 1.1. When R does not contain a field, the algebra (5) and Lemma 2.2 give the result with $u = p$ and $v = p - a$. In the remaining case, Lemma 2.1 allows $T = R[[t]]$ and $u = v = t$. This ring is complete of dimension one, t is regular and outside its maximal-ideal square, and an Artinian ring is Cohen–Macaulay. □

3 Transfer of categorical generation

We recall the exact deformation properties needed. If $(A, \mathfrak{a}, k)$ is local and $x \in \mathfrak{a}$ is A-regular, then

$$A/(x) \text{ dominant} \implies A \text{ dominant}. \tag{8}$$

The converse holds when $x \notin \mathfrak{a}^2$ [6, Theorem 5.6].

For quantitative generation, let $\langle X \rangle_j$ be the subcategory obtained from shifts, finite direct sums, and direct summands of X using at most $j-1$ successive extensions, with $\langle X \rangle_0 = \{0\}$. Takahashi's dominant index is

$$\operatorname{dx}(A) = \inf\{r \in \mathbb{Z}_{\geq -1} : k \in \langle X \rangle_{r+1} \text{ for every } 0 \neq X \in \mathsf{D}_{\mathrm{sg}}(A)\}.$$

A ring is *uniformly dominant* when this index is finite. A regular ring has index -1. For a regular element x, the known bounds are

$$\operatorname{dx}(A) \leq 2\operatorname{dx}(A/(x)) + 1, \qquad \operatorname{dx}(A/(x)) \leq 2\operatorname{dx}(A) + 1 \quad \text{if } x \notin \mathfrak{a}^2; \tag{9}$$

see [7, Theorem 6.2].

Proof of Theorem 1.2. Choose T, u, v from Theorem 1.1. Apply (8) and its converse to each of u, v. This gives

$$G \text{ dominant} \iff T \text{ dominant} \iff R \text{ dominant}.$$

Applying (9) twice gives

$$\operatorname{dx}(R) \leq 2\operatorname{dx}(T) + 1 \leq 4\operatorname{dx}(G) + 3.$$

Interchanging u and v gives the other inequality and the equivalence of finiteness. No uniform bound is inferred from dominance alone. □

4 Gorenstein rings of almost minimal multiplicity

We first verify the hypotheses needed to apply Kimura's theorem to G.

Lemma 4.1. *Let $(R, \mathfrak{m}, k)$ be Gorenstein, with $\mathfrak{m}^3 = 0$, and suppose R is not a complete intersection. Then $G = \operatorname{gr}_{\mathfrak{m}} R$ is a Gorenstein local k-algebra, is not a complete intersection, and has embedding dimension at least three.*

Proof. By [3, Lemma 7.2(1)], $e = \operatorname{edim} R \geq 3$ and $\operatorname{Soc} R = \mathfrak{m}^2 \cong k$. Choose a generator z of $\mathfrak{m}^2$. The multiplication pairing on $\mathfrak{m}/\mathfrak{m}^2$ with values in kz is nonsingular. Indeed, a vector in its radical lifts to an element of $\mathfrak{m}$ annihilating $\mathfrak{m}$, which lies in $\operatorname{Soc} R = \mathfrak{m}^2$ and hence has zero degree-one class. Thus G has socle kz, and is Gorenstein with Hilbert function $(1, e, 1)$.

Write $G = k[[Y_1, \ldots, Y_e]]/I$ in its graded presentation. The space of quadratic relations has dimension $\binom{e+1}{2} - 1$. As I contains no linear form, these relations give linearly independent classes in $I/(Y_1, \ldots, Y_e)I$. Therefore

$$\mu(I) \geq \binom{e+1}{2} - 1 > e \qquad (e \geq 3).$$

An Artinian complete intersection in this regular local ring has a defining ideal minimally generated by e elements. Hence G is not a complete intersection. □

Proof of Theorem 1.3. If $\mathfrak{m}^3 = 0$, Lemma 4.1 makes G a non-complete-intersection Gorenstein local ring containing a field. Kimura's theorem [2, Theorem 7.3(1)] says that G is dominant. Theorem 1.2 now implies that R is dominant.

Suppose $e(R) \le \operatorname{codim} R+2$ and the residue field is infinite. Set $d = \dim R$. Choose $x_1, \dots, x_d \in \mathfrak{m}$ whose initial linear forms give a homogeneous system of parameters for $\operatorname{gr}_{\mathfrak{m}} R$. This is possible over an infinite field. Their ideal Q is a minimal reduction of $\mathfrak{m}$, and their classes in $\mathfrak{m}/\mathfrak{m}^2$ are linearly independent. Since R is Cohen–Macaulay, this is a regular parameter sequence. The Artinian quotient $A = R/Q$ therefore satisfies

$$\ell_A(A) = e(R), \qquad \operatorname{edim} A = \operatorname{edim} R - d = \operatorname{codim} R.$$

These are the usual minimal-reduction equalities, also used in [2, proof of Theorem 7.3(2)]. The ring A is Gorenstein and not a complete intersection: each of these properties is equivalent across quotient by a regular sequence. If $\mathfrak{a}$ is its maximal ideal, then

$$\ell_A(\mathfrak{a}^2) = \ell_A(A) - 1 - \operatorname{edim} A \le 1.$$

Thus $\mathfrak{a}^2$ is zero or a simple A-module, and $\mathfrak{a}^3 = 0$. The Artinian case proves that A is dominant. Applying (8) successively to the parameter sequence proves dominance of R.

For a finite residue field, put $S = R[t]_{\mathfrak{m}R[t]}$. The map $R \to S$ is faithfully flat and local, its maximal ideal is $\mathfrak{m}S$, and its residue field is the infinite field $k(t)$. Flatness gives

$$(\mathfrak{m}S)^j/(\mathfrak{m}S)^{j+1} \cong (\mathfrak{m}^j/\mathfrak{m}^{j+1}) \otimes_k k(t) \qquad (j \ge 0).$$

Consequently the Hilbert functions, embedding dimensions, dimensions, and multiplicities agree. The closed fiber is a field, so S is Gorenstein, and S is a complete intersection if and only if R is; see [5, Tags 0BJL and 09Q7]. The infinite-field case applies to S. Liu's flat-descent theorem [4, Theorem 3.8] then gives dominance of R. $\square$

5 Examples, verification, and scope

The two positions of the residue characteristic in the maximal-ideal filtration both occur. Let $V = \mathbb{Z}_p$, let $e \ge 3$, and form the algebra with basis $1, X_1, \dots, X_e, Z$ and rules

$$X_i X_j = \delta_{ij} Z, \qquad X_i Z = Z^2 = 0.$$

The quotients $T/(p-Z)$ and $T/(p-X_1)$ have Hilbert function $(1, e, 1)$ and nonsingular multiplication pairing. They are Gorenstein and dominant. In the first, p generates the socle and the characteristic is p^2. In the second, p has nonzero degree-one class, $p^2 = Z \neq 0$, and the characteristic is p^3. Their common graded counterpart has characteristic p. These examples illustrate the transfer; the theorem covers arbitrary residue fields and arbitrary symmetric multiplication tensors.

The accompanying standard-library Python program constructs exact finite quotients using the lattice generated by the columns of multiplication by $p - a$. Its triangular matrix has diagonal entries p, giving unique representatives with every coordinate in $\{0, \dots, p-1\}$. The program checks associativity on the integral basis, stability of the relation lattice under multiplication, the quotient cardinality, the first two maximal-ideal layers, the socle, and the position and order of p.

The recorded run covers 258 quotient rings, including 86 Gorenstein and 172 non-Gorenstein cases. It enumerates 16,860 elements and checks 31,698 basis associativity identities. The examples have characteristics $4, 8, 9, 27$. These are diagnostics for the construction; the proof over an arbitrary Cohen ring is the argument of Section 2.

The construction uses $\mathfrak{m}^3 = 0$ at a precise point: every triple product of positive-degree basis elements vanishes, so arbitrary lifts of the multiplication coefficients remain associative. For longer local rings, lifting a multiplication table introduces additional associativity conditions. No corresponding arbitrary-length lifting theorem is asserted. Nor does Theorem 1.2 imply that every non-Gorenstein short ring is dominant.

The quantitative transfer leaves a focused further problem: determine whether the non-complete-intersection Gorenstein graded rings with Hilbert function $(1, e, 1)$ are uniformly dominant, and, if so, bound their dominant indices in terms of e. Any such bound transfers immediately to short Gorenstein rings without a coefficient field by (2).

Preparation. This manuscript was developed with AI assistance. The proof and accompanying computations have undergone internal checks; independent expert review has not been completed.

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